

1. Firstname, lastname, Student ID:
2. Firstname, lastname, Student ID:

INDR 252 CLASS EXERCISE, February 26, 2026

1. Let X_1 and X_2 be two independent and identical random variables with mean μ and standard deviation σ . We define the following random variables:

$$\hat{\theta}_1 = \frac{2X_1 + X_2}{4}$$

$$\hat{\theta}_2 = \frac{X_1 + 3X_2}{4}$$

- a) Find the expected value of both $\hat{\theta}_1$ and $\hat{\theta}_2$ in terms of mean μ and standard deviation σ .
- b) Find the variance of both $\hat{\theta}_1$ and $\hat{\theta}_2$ in terms of mean μ and standard deviation σ .
- c) If both $\hat{\theta}_1$ and $\hat{\theta}_2$ are used as estimators of μ , are they biased or unbiased estimators? Why or why not? Explain.
- d) If you are supposed to choose one of them as a better estimator of μ , which one will you choose? Why or why not? Explain.

$$a) E[\hat{\theta}_1] = E\left[\frac{2X_1 + X_2}{4}\right] = \frac{1}{2} E[X_1] + \frac{1}{4} E[X_2] = \frac{2\mu + \mu}{4} = \frac{3}{4}\mu$$

$$E[\hat{\theta}_2] = E\left[\frac{X_1 + 3X_2}{4}\right] = \frac{1}{4} E[X_1] + \frac{3}{4} E[X_2] = \frac{\mu + 3\mu}{4} = \mu$$

$$b) \text{Var}(\hat{\theta}_1) = \text{Var}\left(\frac{2X_1 + X_2}{4}\right) = \frac{1}{16} (4 \text{Var}(X_1) + \text{Var}(X_2))$$

X_1 & X_2 : independent

$$= \frac{4 \cdot \sigma^2 + \sigma^2}{16} = \frac{5}{16} \sigma^2$$

$$\text{Var}(\hat{\theta}_2) = \text{Var}\left(\frac{X_1 + 3X_2}{4}\right) = \frac{1}{16} (\text{Var}(X_1) + 9 \text{Var}(X_2)) = \frac{10}{16} \sigma^2$$

$$c) E[\hat{\theta}_1] = \frac{3}{4}\mu < \mu \rightarrow \hat{\theta}_1 \text{ is a biased estimator}$$

It is systematically underestimates μ by $\frac{\mu}{4}$ ($= \frac{3}{4}\mu - \mu = -\frac{1}{4}\mu$).

$E[\hat{\theta}_2] = \mu \rightarrow \hat{\theta}_2$ is an unbiased estimator.

d) We choose the estimator with a lower mean square error (MSE).

$$\text{MSE}(\hat{\theta}_1) = (\text{Bias}(\hat{\theta}_1))^2 + \sigma_{\hat{\theta}_1}^2 = \left(\frac{\mu}{4}\right)^2 + \frac{5}{16} \sigma^2$$

$$\text{MSE}(\hat{\theta}_2) = 0 + \frac{10}{16} \sigma^2$$

$$\left(\text{MSE}(\hat{\theta}_1) \leq \text{MSE}(\hat{\theta}_2) \right) \Leftrightarrow \frac{\mu^2}{16} + \frac{5}{16} \sigma^2 \leq \frac{10}{16} \sigma^2$$

$$\Leftrightarrow \mu^2 \leq 5 \sigma^2$$

$$\Leftrightarrow \frac{\mu^2}{\sigma^2} \leq 5$$

Hence, choose $\hat{\theta}_1$ if μ is low enough with respect to

σ , i.e., $\mu^2 \leq 5 \sigma^2$. Choose $\hat{\theta}_2$ if σ is low enough

with respect to μ , i.e., $\sigma^2 \leq \frac{\mu^2}{5}$.